INCH-POUND

MIL-S-44093B(GL)
8 November 1990
SUPERSEDING
MIL-S-44093A(GL)
7 August 1984

MILITARY SPECIFICATION

SHIRT, WOMAN'S, LONG SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a polyester/cotton broadcloth durable press shirt.

1.2 Classification. The shirt shall be of the following types and sizes as specified (see 6.2).

Type I - Standard Neck
Type II - Smaller Neck

Schedule of sizes types I & II (inches)

Size	Sleeve lengths		
	Short	Regular	Long
2	22-1/2	23-1/2	25
4	22-1/2	23-1/2	25
6	22-3/4	23-3/4	25-1/4
8	23	24	25-1/2
	24	25	

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8410

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Schedule of sizes types I & II (inches) (cont'd)

Size	Sleeve lengths		
	Short	Regular	Long
10	23-1/4	24-1/4	25-3/4
	24-1/4	25-1/4	
12	23-1/4	24-1/4	25-3/4
	24-1/4	25-1/4	26-3/4
14	23-1/2	24-1/2	26
	24-1/2	25-1/2	
16	23-3/4	24-3/4	26-1/4
	24-3/4	25-3/4	
18	23-3/4	24-3/4	26-1/4
		25-3/4	
20	24	25	26-1/2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
A-A-50199	-	Thread, Polyester Core, Cotton- or Polyester-Covered
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-F-21840	-	Fastener Tape, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For
MIL-T-43624	-	Thread, Polyester, Spun
MIL-C-43992	-	Cloth, Broadcloth, Polyester/Cotton, Durable Press
MIL-T-44193	-	Thread, Polyester, Textured

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MIL-C-44296 - Cloth. Fusibles
MIL-P-55010 - Plastic Sheet, Polyethylene Terephthalate

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Technical Manual of the American Association of Textile Chemists and Colorists

Method 88B (two photographs) - Photographic Comparative Rating of Single and Double Needle Seams

Method 124-82 - Appearance of Durable Press Fabrics After Repeated Home Launderings

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709-2215.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Type 304 - Stainless and Heat Resistant Steel
Type 316 - Stainless and Heat Resistant Steel

(Application for copies of Steel Product Manual should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-204 - Methods of Testing Sewing Threads
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

Department of Defense Standard Shades for Buttons

(Application for color cards should be addressed to the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material shall be polyester/cotton broadcloth, Army Green 415, conforming to MIL-C-43992.

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3.3.2 Fusible interlining.

3.3.2.1 Shoulder loops, cuffs, top collarstand (alternate method) and front facing. The fusible interlining material for the shoulder loops, cuffs, top collarstand (alternate method), and front facing shall conform to type V, style B, class 2 of MIL-C-44296.

3.3.2.2 Top collar. The fusible interlining for the top collar shall conform to type I, style D, class 3 of MIL-C-44296. As an alternate, the fusible interlining as described in 3.3.2.1 may be used.

3.3.2.3 Collarstand (automatic creaseband construction) fusible. The fusible for the collarstand shall be 100 percent cotton with a minimum weight of 1.8 ounces per square yard and 44 warp yarns per inch minimum and 34 filling yarns per inch minimum. As an alternate, a 100 percent thermal bonded nonwoven weighing a minimum of 1.5 ounces per square yard may be used. The fusible shall be treated with polyvinyl copolymer thermoplastic or ethylene vinyl acetate dot adhesive on each side and be transitory in nature such that the adhesive will wash out after two launderings in accordance with labeled instructions. The base fusible material shall be white and shrink resistant as not to exceed 2.0 percent shrinkage in either warp (machine) and filling (cross - machine) direction (see 3.10).

3.3.2.4 Fusing conditions. A single layer of fusible interlining shall be fused to a single layer of basic material on an electric dry heated conveyor or flathead type fusing press which has the capability of controlling and retaining pressure, dwell time, and temperature within an 8-hour period. For optimum results with a particular fusible style and press used, setting should be obtained from the manufacturer of the fusible interlining.

3.3.2.5 Bonding strength. The fusibles in 3.3.2.1 and 3.3.2.2 shall meet the initial and laundered bonding strength requirements as specified in MIL-C-44296 when fused to the basic material.

3.3.3 Labels. Each shirt shall have a size label and a combination identification-instruction label conforming to DDD-L-20. The fastness to launder-ing requirements of DDD-L-20 shall apply. The size label shall conform to type IV or VI, class 2, except that "Type I - Standard neck" or "Type II-Smaller neck", as applicable, must appear on size labels. The combination identification-instruction label shall conform to type VI, class 15, except that the label cloth shall be dyed Lemon Yellow Cable No. 7020, in conformance with the Standard Color Card of America. The width of the combination label shall be not more than 1-3/8 inches and the length shall be governed by the contents. The combination label shall contain the following information:

SHIRT, WOMAN'S, LONG SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

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Washing instructions

1. Wash with moderately hot water (115°F) using soap or mild detergent.

   a. For machine washing. Use wash and wear cycle.
   b. For hand washing. Do not wring or twist.

2. DO NOT STARCH OR BLEACH.

3. Dry in tumble dryer at warm setting. Remove immediately at end of drying. Shape and hang.

4. To drip dry, remove from water before last spin cycle. Shape and hang.

5. Touch up is recommended. Iron at a moderate or synthetic material heat setting.

DO NOT REMOVE THIS LABEL

3.3.4 Thread. The thread for seaming, stitching, buttonholes and button sewing shall be cotton- or polyester-covered polyester core, ticket No. 70, 2 or 3 ply and 100, 2 or 3 ply conforming to A-A-50199. The thread shall be stabilized so as not to shrink in boiling water more than 1.5 percent when tested as specified in table IV. Spun polyester thread conforming to class 1 or 2 of MIL-T-43624 may be substituted for cotton- or polyester-covered polyester core thread as follows:

Covered polyester	Spun polyester
70, 2 or 3 ply	70, 2 or 3 ply
100, 2 or 3 ply	100, 2 ply

As an option, for all stitching specified in table II, size 100 thread may be used for all operations except for sewing on buttons and making buttonholes. As another option, the looper thread for overedging the front facing fusible interlining to the front edge of the basic material may be polyester textured, type I, class 2, size 18 Tex, conforming to MIL-T-44193.

3.3.4.1 Color and colorfastness. The thread shall be White AH, C.A. 66050. The dyed thread shall show fastness to light, perspiration, and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light, perspiration, and laundering.

3.3.5 Buttons. The buttons shall conform to type II, class K, style 25, 19 line of V-B-871. The color of the buttons shall be shade White BA, C.A. 62031.

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3.3.6 Collar stays. The collar stays shall be made from polyethylene terephthalate plastic sheet conforming to MIL-P-55010, except the thickness shall be 0.010 ± 0.001 inch, the color shall be natural (clear or opaque), and the transparency and optical properties shall not apply. Either both ends shall be rounded or pointed with the points rounded, or one end rounded and one end cut at an angle with the point rounded (see 4.4.1.2).

3.3.6.1 Collar stay configuration.

3.3.6.1.1 Unslotted stays. The stays shall measure 3/8 ± 1/32 inch wide and 2-1/4 ± 1/16 inches long.

3.3.6.1.2 Slotted stays. The stays shall measure 3/8 ± 1/32 inch wide and 2-1/4 ± 1/16 inches long. The stay shall have two slots 3/4 ± 1/16 inch long, centered along the length of the stay. Each slot shall be 1/32 inch wide and spaced approximately equidistant across the width of the stay.

3.3.7 Fastener tape, nylon. The nylon tape fastener shall be 5/8 inch wide, conforming to type I, class 1, of MIL-F-21840. The color for the neckband loop fastener tape shall be Gray 3123.

3.4 Design The shirt is a semi-fitted hip length overblouse style with long sleeves, shoulder loops, collarstand, 6 button cut away front, side body, and darts at waist, bust and back neck. The collar, collarstand, shoulder loops, cuffs, and front facing have a fusible interlining and the collar has stays. Loop fastener tape is attached to stand for the attachment of a black necktab (see figure 1).

3.5 Figure. Figure 1 is furnished for information purposes only. If there are any inconsistencies between the specification and the figure the specification shall control.

3.6 Patterns. Standard patterns will be furnished by the Government. The seam allowance for the cuff joining seam shall be 3/8 inch. All other seams shall be 1/4 inch unless otherwise specified. The standard patterns shall not be altered in any way (except when automated apparel equipment is used for either making or setting cuffs, collar, collarstand, and/or shoulder loops see 3.10) and are to be used on as a guide for cutting the contractor's working patterns, which shall be identical to the standard patterns. In the safety stitch construction the 1/4 inch seam allowance shall be maintained even though the finished seam allowance shall measure 3/16 to 7/32 inch due to trimming of the ravelled edges of fabric.

3.6.1 List of pattern parts. The component parts of the shirt shall be cut from material as specified in accordance with the pattern parts indicated.

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TABLE I. List of pattern parts

Material	Nomenclature of pattern parts	Cut parts
Basic material	Front	
	Type I	2
	Type II	2
	Back	
	Type I	1
	Type II	1
	Side body	
	Type I	2
	Type II	2
	Sleeve	2
	Sleeve binding	2
	Cuff	2 1/
	Top collar	
	Type I	1
	Type II	1
	Under collar	
	Type I	1
	Type II	1
	Collar stand	
	Type I	2
	Type II	2
	Stay holder 2/	
	Type I	2
	Type II	2
	Shoulder loop	4
Fusible interlining	Collar stand fusible interlining	
	Type I	1
	Type II	1
	Top collar fusible interlining	
	Type I	1
	Type II	1
	Shoulder loop fusible interlining	2
	Cuff fusible interlining	2 1/
	Front facing fusible interlining	
	Type I	2
	Type II	2

1/ When cuff setting alternate operation 22.b. is used, add 1/8 inch to the cuff and cuff fusible interlining patterns.

2/ If used.

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3.7 End item seam appearance, bond strength of fusibles and adhesion of ultra-sonically fused collar stays.

3.7.1 Seam appearance rating. After the finished shirt has been laundered five times, the appearance rating of the seams and stitching shall be as follows (see 4.4.5):

Part	Minimum lot average (3 observations per sample unit)
Fused collar	3.8
Fused shoulder loops	3.8
Fused cuffs	3.8
Armholes	2.8
Bottom hem	2.8
Front-waist darts	2.8
Bust darts	2.8
Back neck darts	2.8
Front side body seam	2.8
Back side body seam	2.8

3.7.2 Appearance of fusibles. After the shirt has been laundered ten times, the shirt shall show no bubbling or delamination of the fused plies of material when tested as specified in 4.4.5. There shall be no showing of the "strike-through" of the adhesive after fusing.

3.7.3 Ultrasonically fused collar stay resistance to laundering (when applicable). After the shirt has been laundered ten times, the shirt shall show no delamination of the ultrasonically fused collar stays when tested as specified in 4.4.5.

3.8 Construction. The construction shall conform in all respects to the requirements specified in table II and herein.

3.8.1 Stitches, seams and stitching. Stitch, seam and stitching types specified in table II shall conform to FED-STD-751. Where two or more seam or stitch types are given for the same part of an operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the shirt. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained (see 6.5), so that there is no tight or loose stitching. The gage of the safety stitch seams shall be 1/4 inch. The guides and knives on safety stitch machines shall be set to trim only the ravelled ends of the fabric. The edge and top stitching shall be uniformly gaged. Seams required to be worked out, that have a depth between the fold of 1/16 inch or more shall be considered a defect.

3.8.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table II.

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3.8.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be back-tacked 1/4 to 1/2 inch. Ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch. The ends of the label stitching shall be overlapped not less than three stitches. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired by using 301 stitch type. The series 500 stitch types for sleeve setting seams shall be run off with a 1-1/8 $\pm$ 1/8 inch tail.

3.8.4 Buttonholes. The buttonholes shall be straight-cut, purl or whip type and the finished inside cut length shall be 1/2 to 9/16 inch long. The stitching shall be securely caught in the fabric and the ends shall be securely tacked. When a purl-type buttonhole is used, the purling shall be on the outside of the shirt. All loose thread ends shall be trimmed or fused to within 1/8 inch of the tack. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

3.8.5 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three or more tying, overlapping or backstitches are used to secure the ends of the stitching.

3.8.6 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch.

3.8.7 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the shirts.

3.9 Manufacturing operations requirements. The shirt shall be made in accordance with the operation requirements specified in table II. The contractor is not required to follow exact sequence of operations listed unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished shirt.

3.10 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table II, providing that the seam and stitch type are as specified and the finished component conforms to the required size dimension and configurations. When a government furnished shaper pattern is forwarded, caution shall be taken to assure the component conforms to that pattern. Patterns may be altered to accommodate the use of automated apparel equipment as specified in 3.6.

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TABLE II.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut shirts in strict accordance with patterns furnished which show size, directional lines, marks, and notches for proper assembly. Directional lines shall be placed in the warp direction. The use of drill holes is permitted for the darts and sewn labels, providing the drill holes are covered by the outer fabric or label, respectively. Otherwise drill holes shall be considered to be holes in the fabric. b. Cut component parts from the same piece of the basic material, except the undercollar, stay holder, sleeve binding, and the under parts of collarstand which may be cut from ends. Parts cut from ends shall approximate the shade of the shirt. The binding for sleeve opening may be roll cut providing that the finished width of the binding conforms to the measurements cited in operation 11.a. c. Cut the fusible interlinings in accordance with directional lines in patterns.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3.1 shall be removed from production and replaced with non-defective and properly matched components.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Marking and fusing. a. Mark, ticket or bundle all component parts of the shirt except those parts cut from ends to insure correct size, proper assembly and a uniform shade throughout the shirt. b. Any method of marking may be used except: (1) Metal fastening devices. (2) Sew-on tickets. (3) Adhesive type tickets which leave traces of adhesive on the material after removal of the tickets. c. The use of ink pad numbering machine, rubber stamp or pencil will be acceptable provided the numbers do not show through the outside of the shirt and the numbers are covered by the seam allowance. d. Fuse the following small parts (see 3.3.2): (1) Cuffs. Place fusible even with foldline. All remaining edges shall be set back 1/8 inch. Fuse top cuff only. (2) Top collar and shoulder loops. Position in center with edges set back 1/8 inch. Fuse top collar and top shoulder loop only. (3) Collarstand. Position in center with all edges even. Fuse top collarstand only (for alternate method only).						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labeling.						
	Center size label (type IV) on top collarstand (1/4 inch off center tolerance), so that on the finished shirt, the bottom of the size label (type IV) will be not less than 1/4 inch or more than 1/2 inch from the seam joining the collarstand to the neck. (see operation 19.a. if type VI label is used.)						
5.	Make collar.						
	a. Position stay holder to undercollar with side and top edges even.						
	b. Form a tunnel for the collar stays by stitching the stay holder to undercollar with two rows of stitching 7/16 to 1/2 inch apart according to marks on pattern. The collar points shall be aligned with the center of the two rows of stitching and the stitching shall extend to the edges of the stay holder.	301	SSv-2	9-12	70	70	
	c. Insert collar stays into the tunnels with rounded ends abutting the collar and undercollar joining seam and stitch along bottom stay holder from top to bottom edges of collar.	301	SSv-1	9-12	70	70	
	or						
	When using slotted collar stays, position collar stays on undercollar according to marks on pattern so that tapered ends are 3/16 inch back of undercollar point joining seam. Stitch collar stays to undercollar with a 3/4 inch long tack. The stitching shall straddle the center bar of the stay and not penetrate the plastic material.	101		20-24	70		

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar. (cont'd)						
	or						
	Position collar stays according to marks on pattern so that tapered ends are 3/16 inch back of under-collar point joining seam. Ultrasonically fused the collar stay to the undercollar with two rows of fusing points the full length of the stay. Each row shall contain nine to eleven fusing points per inch.						
	d. With the fused top collar and undercollar face to face, stitch around side and top edges.	301 or 401	SSe-2(a) SSe-2(a)	9-12 9-12	70 70	70 100	
	e. Trim, turn and press collar points on a collar pointing machine.						
	f. Edge stitch top and sides of collar 1/4 ± 1/16 inch from edge.	301	SSe-2(b)	9-12	70	70	
6.	Make collarstand and attach collar.						
	a. Place collar between the fused top collarstand and under collarstand (so that the fused top collar-stand and under collar are face to face) as indicated by notches on pattern and stitch from end to end a-long top edge of collarstand, 3/16 to 1/4 inch from edge. Turn top and under collarstand to finished position and raise stitch 1/16 to 3/32 inch from turned edge to ends of collar. This stitching may continue around the ends of the collarstand (see operation 19.e.).	301 or 401 and 301	SSq-2(a) SSq-2(a) SSq-2(b)	9-12 9-12 9-12	70 70 70	70 100 70	
	or						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make collarstand and attach collar. (cont'd)						
	b. Automatic creaseband construction (when used). Superimpose collarstand interlining (see 3.3.2.3) on both top and under collarstand. An automatic creaseband machine may be used, which folds the edges of the collarstands and interlinings simultaneously. The interlinings are held in place by the activated adhesive of the fusible interlinings. When this method is used, operation 19.c will be deleted.						
	c. Automatic profile stitch machine construction (when used). Operation 6.a may be accomplished with an automatic profile stitch machine which places the collar between the previously creased and tacked collarstands and stitches along the top edge of the collarstand 1/16 inch from edge.						
7.	Make cuffs.						
	Finished appearance. The buttonholes shall be clean cut with the stitching securely caught in the fabric. The cuffs shall finish 1-3/4 ± 1/8 inches wide and have square corners.						
	a. Fold cuff lengthwise and stitch ends. Turn and force out edges and corners.	301 or 401	SSe-2(a) SSe-2(a)	9-12 9-12	70 70	70 100	
	b. Make a horizontal buttonhole in center of each cuff (1/8 inch off center tolerance permitted) with the inside cut end of the buttonhole 3/8 ± 1/16 inch from front edge (see 3.8.4).	304	Whip or purl types	42-52 per buttonhole including tack	70	70	
	c. Press cuffs on a hot head pressing machine (see operation 26).						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make shoulder loops. Finished appearance. The shoulder loops shall finish smooth and flat and uniform in construction and place ment. The buttonholes shall be clean cut and correctly positioned. The shoulder loops shall measure finished 1-5/8 ± 1/8 inches at the narrow end and 1-7/8 ± 1/8 inches at the wide end.						
	a. Position the shoulder loops face to face and stitch around all edges except armhole end.	301 or 401	SSe-2(a) SSe-2(a)	9-12 9-12	70 70	70 100	
	b. Turn loop, work out points and edges and edge stitch along sides and point, 1/16 to 3/32 inches from edge.	301	SSe-2(b)	9-12	70	70	
	c. Make a horizontal buttonhole in the center of each loop 1/8 inch off-center tolerance, with the end of the buttonhole 3/8 to 7/16 inch from point of loop (see 3.8.4).	304	Whip or purl types	42-52 per buttonhole including tack	70	70	
	d. Press shoulder loops with a hot head pressing machine (see operation 26).						
9.	Attach facing interlining to fronts, and label to facing.						
	a. Superimpose fusible interlining (adhesive side up) to front facing, with front edges even overedge the fusible interlining to the front of the shirt.	504	SSa-1	14-16	100-2	100-2 or 18 TEX	
	b. Fold back the facing as indicated by patterns and fuse facing to the shirt body.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Attach facing interlining to fronts, and label to facing. (cont'd)						
	NOTE: The line of the overedge stitching shall not be visible or cause any impressions on the outside of the shirt after fusing.						
	c. Position the combination identification and instruction label on the left facing (AS WORN), 3 to 3-1/4 inches from finished hem and 1-1/6 to 1-3/16 inch from finished front and stitch on all four sides. The stitching shall not be through the printing.	301	LSbj-1	6-8	70	70	
10.	Make sleeve.						
	Join sleeve underseam together with safety stitch.	515 or 516 or 519	SSa-2	9-12	70 (chain and 100	100 100 (over-edge)	
11.	Make sleeve openings and fold pleats.						
	Finished appearance. The binding of the sleeve opening shall be uniform in width and shall finish smooth and flat. Sleeve openings shall measure 3 ± 1/8 inches along outside fold exclusive of cuff.						
	a. Turn raw edges of the binding to the inside, insert the edges of sleeve opening 1/4 ± 1/16 inch into binding and seam. The binding shall finish 3/8 to 7/16 inch wide.	301 or 401	BSc-1 BSc-1	9-12 9-12	70 70	70 100	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make sleeve openings and fold pleats. (cont'd)						
	b. Turn binding to inside of sleeve and stitch or bartack diagonally through binding only. An automatic straight tack may be used.	301 or Bartack		9-12 28 per bartack	70 70	70 70	
	c. Fold pleats at bottom edge of sleeve as indicated by notches and stitch 1/16 to 1/8 inch from raw edge. The folded edges of the pleats on the outside shall face towards the back of the sleeve opening.	301	OSf-1	9-12	70	70	
12.	Make bust and front-waist darts.						
	Finished appearance. The darts shall be properly positioned and finished without pleats, puckers or gathers.						
	a. Fold and stitch bust darts in fronts tapering to nothing 1/2 to 5/8 inch beyond drill hole. The darts shall be neatly formed and tapered to a flat point. The point shall be securely tacked or backstitched.	301	OSf-1	9-12	70	70	
	b. Press bust dart smooth and flat with a heated pressing iron or hot head pressing machine (see operation 26). The edges of the bust shall be shall be pressed down.						
	c. Notch corner of front-waist dart and join seams, tapering to nothing 1/2 to 5/8 inch beyond mark for drill hole on pattern.	301	SSa-1	9-12	70	100	
	d. Overedge stitch the front waist dart seams catching both plies of fabric.	502 or 503	EFd EFd	9-12 9-12	70 70	70 70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Make bust and front-waist darts. (cont'd)						
	e. Press front-waist darts smooth and flat with a heating pressing iron or hot head pressing machine (see operation 26). The edges of the darts shall be pressed towards the front. Top stitch front-waist darts at the curve (where dart goes from vertical towards side seam) 1/16 to 3/32 inch from folded edge. Stitching shall be 1/2 to 5/8 inch in each direction. Back stitching is required.	301	LSbm-3	9-12	70 (Pale Green)	70 (Pale Green)	
13.	Make back.						
	a. Fold and stitch back darts tapering to nothing 1/2 to 5/8 inch beyond mark for drill hole on pattern. The darts shall be neatly formed and tapered to a flat point. The point shall be securely tacked or backstitched.	301	OSf-1	9-12	70	70	
	b. Press back darts smooth and flat with a heated pressing iron or hot head pressing machine (see operation 26). The folded edges of the darts shall be pressed toward the center back.						
	a. Join side seams of front to front seam of side body with safety stitch, matching notches and easing front to side body between notches in bust area.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 / 100 (over-edge)	
	b. Press seam smooth and flat with a hot head pressing machine (see operation 26). The folded edges of the seam shall be pressed toward the front.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Join shoulder seams and back to fronts.						
	a. Join shoulder seams of front and back (top of facing shall not be caught in shoulder seam) with safety stitch extending to overedged stitched edge of facing. Ease back shoulder seam to front shoulder seam between notches and press seam toward back.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (over-edge)	
	b. Join back seam of side body to side seam of back with safety stitch matching notches and easing back to side body between notches in shoulder blade area.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (over-edge)	
	c. Press seam smooth and flat with a hot head pressing machine (see operation 26). The folded edges of seam shall be pressed toward the center back.						
16.	Set shoulder loops.						
	Position shoulder loop so point of loop is 1/4 to 3/8 inch forward of shoulder seam, 1/16 to 1/8 inch from collar joining finished seam with front edge of loop at armhole opening 1 $\pm$ 1/16 inch forward of shoulder seam. Stitch 1/8 to 3/16 inch from raw edge of armhole opening and trim excess even with opening. The excess shoulder loop may be trimmed in operation 17. The fused outer edge must be on top.	301 or 401	SSa-1 SSa-1	9-12 9-12	70 70	70 100	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Set sleeves.						
	Position sleeves in armholes with the front notch matching the front seam of side panel and the back notch matching the back seam of side panel and join with safety stitch catching the raw edges of the shoulder loop in the seaming. The sleevehead fullness shall be distributed evenly between the notches at top.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (overedge)	
18.	Join cut-a-way front facing to front together at bottom of overblouse.						
	a. Position the facing and front face to face and stitch angular edges together and continue stitching across width of facing bottom.	301	SSa-1	9-12	70	70	
	b. Turn to finished position and press smooth and flat with seamline to underside.						
19.	Join collar to shirt and attach size label.						
	Finished appearance. The collar shall be carefully joined to the shirt with notches matching without twists, puckers or distorting the collar or shirt. The collar ends shall not be uneven in length by more than 1/8 inch. In stitching down collarstand, the stitching shall not be more than 1/8 inch on or off the opposite side.						
	a. Position size label at center back and stitch 1/8 to 3/16 inch from raw edge of neck area. (Type VI label, if applicable).	301	SSa-1	9-12	70	70	
	b. Fold right and left front facings of shirt to the inside along center front fold line.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Join collar to shirt and attach size label. (cont'd)						
	c. Turn in the bottom edge of the fused top collarstand and stitch 1/16 to 3/32 inch from turned edge.	301	EFa-1	9-12	70	70	
	d. Stitch under collarstand to inside of shirt with seam allowance turned toward stand. The right and left facing folded edges shall finish even with ends of collarstand.	301	LScf-2(a)	9-12	70	70	
	e. Stitch down the fused top collarstand 1/16 to 3/32 inch from turned edge through stand and shirt; continue the stitching around the ends of the collarstand if not previously stitched (see operation 6.b.).	301	LScf-2(b)	9-12	70	70	
20.	Join shoulder end of facing to shoulder seam.						
	Stitch shoulder end of facing to shoulder seam edge, 1/16 to 1/8 inch from edge, for a distance of 3/4 to 1 inch from outside edge of facing. Facing shall lay smooth and flat without twists or puckers.	301	SSa-1	9-12	70	70	
21.	Hem bottom.						
	a. Turn under bottom raw edge of hem 1/4 inch and stitch 1/16 to 1/8 inch from folded edge.	301	EFa-1	9-12	70	70	
	b. Turn up hem according to notches on pattern and tack back edge of facing to hem not less than 1/2 inch along overedge of facing.	301	SSa-1	9-12	70	70	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Hem bottom. (cont'd)						
	c. Blindstitch hem partially penetrating body fabric. The stitching shall extend across the front facings, 3/4 to 1 inch from overedge stitching edge. The front side body seams shall turn toward the front and the back side body seams shall turn toward the back. The finished hem shall measure 1-3/2 inches wide.	103 or 106	EF1-1	3-4	70 (Pale Green)		
22.	Join cuffs to sleeve.						
	a. Seam the inside of the under cuff and interlining to bottom of sleeve on the buttonhole, with the bound edge of the sleeve opening turned to the inside at buttonhole end of cuff and caught in the stitching.	301 or 401	LScf-2(a) LScf-2(a)	9-12 9-12	70 70	70 100	
	b. Turn in sleeve and cuff seam allowance towards cuff. Turn in fused top cuff and stitch to the bottom of sleeve on the outside 1/16 to 3/32 inch from edge. Turn corner and continue edge stitching down the side of cuff, along bottom edge and up opposite side.	301	LScf-2(b)	9-12	70	70	
	or						
	(See 1/ and 2/ to table I). Cuffs may be attached in one operation providing the finished appearance is the same as specified in a. and b. above, and the ends of the row of stitching are backtacked or crossed with another line of stitching.	301	LSe-1	9-12	70	70	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
23.	Attach necktab fastener pile tapes.						
	Stitch a 5/8 inch square gray fastener pile tape on the right and left top collarstand with the tape centered on the collarstand. The front edge of tape shall be positioned 1-3/8 ± 1/16 inch from collar edge, where collar is joined to collarstand on each side of shirt. The stitching shall penetrate the undercollar stand.	301	LSbj-1	9-12	70	70	
24.	Make buttonholes.						
	a. Make five vertical 1/2 to 9/16 inch inside cut length buttonholes in the right front of the shirt through the facing centered over marks on pattern positioned 1 ± 1/16 inch in from front edge (see 3.8.4).	304	Whip or purl types	42-52 per buttonhole including tack	70	70	
	b. Make one horizontal 1/2 to 9/16 inch inside cut length buttonhole in the right front of shirt in center of collarstand with front inner cut edge of buttonhole extending 1/16 to 1/8 inch forward of front vertical buttonholes.	304	Whip or purl types	42-52 per buttonhole including tack	70	70	
25.	Clean shirts.						
	a. Before folding and packaging, shirts must have all thread ends trimmed and removed, be free of residual fiber particles, lint and dust, or other foreign substances, and spots and stains removed.						
	b. Remove all shade tickets.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
26.	Pressing. Press the bottom of shirt, side seams, underarm seams and cuffs. NOTE: The use of steam inflated bag type pressing machine is prohibited. The hot head pressing machine cited in operations 7.c., 8.d., 12.b., 12.e., 13.b., 14.b. and 15.c., shall be equipped with controls for accurate temperature (250° to 270°F), time (minimum 10 seconds), and pressure (70 to 80 psi gauge).						
27.	Sew on buttons. Finished appearance. The buttons shall be aligned and securely sewn in the center of the buttonholes ± 1/16 inch in front of shirt, shoulder loops, and cuffs. Collar ends shall be centered equidistant from center of shirt front and shall open 5/8 ± 7/8 inch when collar is buttoned.						
	a. Sew one button centered on left end of collarstand 1-1/8 ± 1/16 inch from front of shirt. When collar is buttoned, center of button shall be in line with vertical buttonholes on front without distorting left or right front.	301 or 101	-	14-16 per button 14-16 per button	70 70	70	
	b. Sew five buttons on left shirt front 1 ± 1/16 inch from the front edge through the front facing.	301 or 101	-	14-16 per button 14-16 per button	70 70	70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
27.	Sew on bottons. (cont'd)						
	c. Sew two buttons on the outside of each cuff. One botton shall be sewn at the back end, to correspond with the bottonhole, with the center of the button 3/8 to 1/2 inch from the edge. The other button shall be positioned in direct line with the first button with the center of the button 1-1/8 to 1-1/4 inches from the edge.	301 or	-	14-16 per button	70	70	
		101	-	14-16 per button	70		
	d. Sew one button on each shoulder through facing to correspond with the buttonhole in the shoulder loop. The shoulder loops shall lay smooth and flat without puckers or twists when they are buttoned.	301 or	-	14-16 per button	70	70	
		101	-	14-16 per button	70		
28.	Touch-up pressing and closing shirt.						
	a. Examine for complete shirt pressing.						
	b. When required, touch-up fronts, backs, cuffs bottom hems and collars, before folding and packaging with a heated iron at a permanent press (moderate) setting.						
	c. Button shirt fully.						

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3.10 Measurements. The finished measurements of the long sleeve shirt shall conform to the applicable measurements shown in table III.

TABLE III. Finished measurements (inches)

SIZE	COLLAR LENGTH 1/		1/2 BUST 2/	CROSSBACK 3/	1/2 SWEEP 4/	BACK LENGTH 5/			SLEEVE LENGTH 6/		
	TYPE I	TYPE II				S	R	L	S	R	L
2	13-5/8	13	17	13-3/4	17-1/2	19-3/8	20-5/8	21-7/8	22-1/2	23-1/2	25
4	13-7/8	13-1/4	17-1/2	14	18	19-5/8	20-7/8	22-1/8	22-1/2	23-1/2	25
6	14-1/8	13-1/2	18	14-1/4	18-1/2	19-7/8	21-1/8	22-3/8	22-3/4	23-3/4	25-1/4
8	14-3/8	13-3/4	18-1/2	14-1/2	19	20-1/8	21-3/8	22-5/8	23 24	24 25	25-1/2
10	14-5/8	14	19	14-3/4	19-1/2	20-3/8	21-4/8	22-7/8	23-1/4 24-1/4	24-1/4 25-1/4	25-3/4
12	14-7/8	14-1/4	19-3/4	15-1/8	20-1/4	20-5/8	21-7/8	23-1/8	23-1/4 24-1/4	24-1/4 25-1/4	25-3/4 26-3/4
14	15-1/8	14-1/2	20-1/2	15-1/2	21	20-7/8	22-1/8	23-3/8	23-1/2 24-1/2	24-1/2 25-1/2	26
16	15-3/8	14-3/4	21-1/4	15-7/8	21-3/4	21-1/8	22-3/8	23-5/8	23-3/4 24-3/4	24-3/4 25-3/4	26-1/4
18	15-5/8	15	22-1/4	16-3/8	22-3/4	21-3/8	22-5/8	23-7/8	23-3/4	24-3/4 25-3/4	26-1/4
20	15-7/8	15-1/4	23-1/4	16-7/8	23-3/4	21-5/8	22-7/8	24-1/8	24	25	26-1/2
TOLERANCE	± 1/4		± 1/2	± 3/8	± 5/8		± 3/8			± 1/4	

1/ Collar measurement taken along center of collarstand, from center of button to front cut end of buttonhole, with collar in a flat position.

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2/ Measure across front 1 inch below armhole seam with shirt buttoned and laid flat.

3/ At bottom ends of darts, measure across back from sleeve setting seam to sleeve setting seam.

4/ Measure sweep along bottom edge from folded edge to folded edge with shirt buttoned.

5/ Back length measurements taken down center back from undercollar seam to bottom of shirt.

6/ Sleeve length measurement taken along outside of sleeve from sleeve setting seam at shoulder to bottom edge of cuff.

3.11 Workmanship. The finished shirts shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on the components listed in table IV for the characteristics noted. Unless otherwise indicated, testing shall be conducted in accordance with FED-STD-191. All test reports contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot size	Sample size (sample units)
Up to 22,000 (inclusive)	8
22,001 and over	13

The basis for lot size and sample units for testing shall be as follows:

Component	Lot size expressed in	Sample unit for testing
Fusible interlining fused to basic cloth	Yards	1 yard full width
Thread	Cones	1 cone

TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method
Fusible interlining fused to basic cloth	Initial adhesion	3.3.2.5	4.4.2.1
	Adhesion after 10 launderings	3.3.2.5	4.4.2.1 1/

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TABLE IV. Component tests (cont'd)

Component	Characteristic	Requirement paragraph	Test method
Thread	Shrinkage	3.3.4	ASTM D 204 2/

1/ Material shall be laundered as specified in 4.5.1.2 prior to testing for adhesion.

2/ Shrinkage in dry heat shall not apply.

4.4.1.2 Examination of collar stays. Eight collar stays shall be randomly selected from each lot and examined for conformance to configuration and dimensional requirements in 3.3.6. Any nonconforming collar stay found in the sample shall be cause for rejecting the lot represented.

4.4.2 In-process inspection. Visual examination shall be accomplished to ascertain that collar stay tunnels are located and constructed as specified, collar stays are inserted into the collar in the position specified, collar stays are attached and located as specified, and that the fusible material is attached to shirt collar and interlining to shirt fronts, collarstand, cuffs and shoulder loops as specified. Evidence of nonconformance shall be cause for rejection of all affected items. Final determination of any suspected nonconformance shall be accomplished at the discretion of the Government by partial or complete disassembly of the end items in question.

4.4.2.1 Adhesion testing of fusibles. Prior to fusing each 1,000 parts, a sample of equivalent area to a part shall be fused for visual and adhesion testing. The fusible shall have a 1 inch starter strip or non-fused area at the top portion of the bond strength samples. This can be accomplished either by placing a strip of paper, folding the fusible onto itself or by placing a thin non-adhesive material between the fusible interlining and outershell before fusing. After visual examination for bubbling, delamination, and bleed through, the sample shall be cut to a 1-inch width and peeled in the warp direction using a spring scale or suitable device capable of measuring the peel strength in ounces. Any sample failing the minimum peel strength requirement specified in 3.3.2.2.3 or any bubbling, delamination or bleed-through shall be cause for rejection of the previous 1,000 fused parts. The rejected lot shall not be used in assembly of the shirts. No fusing shall be performed after the rejection of a 1,000 part lot until the machine has been properly adjusted to produce a passing sample. The rejected lot shall then be replaced with a shade matched lot which has been fused correctly. Rejection of the previously fused parts shall not apply to the first sample tested after machine warm-up. A rejected lot may not be used in the assembly of shirts to be presented to the Government. A rejected lot must be disposed of or set aside with clear markings to avoid accidental use. A rejected lot may be replaced with a lot which has been fused correctly and each part must be properly shade matched to the shirt to which it will be joined. A shaded part defect will be scored on all improperly matched shirts.

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table V. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects, 6.5 for major and minor A combined defects, and 15 for total (major, minor A, and minor B combined) defects.

TABLE V. End item visual defects

Defect	Classification Major	Minor A	Minor B
Pressing			
Omitted, poorly pressed, not pressed as specified exhibiting a double crease line		201	
Burned or scorched	101		
Gloss, wrinkles, or creases			301
Edge bulky (not pressed solidly)			302
Sleeve center creased			303
One or more darts or seams not pressed in the direction specified			304
Cuff not pressed evenly			305
Material defects and damages			
Hole, cut, tear, needle chew, burn, mend, darn, patch, multiple float, jerked-in filling, shade bar, or broken or missing yarn:			
On outside	102		
On inside		202	

NOTE: The terms outside and inside are defined as follows:

Outside: Any part of shirt that is visible when the completely buttoned and closed shirt is examined

Inside: Any part of shirt that is not visible when the completely buttoned and closed shirt is examined

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Material defects and damages (cont'd)			
Slub larger than E in all areas except collar 1/		203	
Slub larger than C in collar 1/	103		
Knots larger than B anywhere on shirt 1/	104		
1/ Only knots and slubs on outside that exceed the limits specified in Sears Fabric Defect Scale (see 6.7) shall be scored as defects in evaluating quality of shirts			
Shaded parts			
Variation in shade within an outside part or between outside parts, except undercollar	105		
Top collarstand, sleeve binding, or undercollar stand, badly shaded		204	
Cleanness			
Any spot or stain: - on outside	106		
- on inside		205	
One or more marking tickets not removed		206	
Any marking clearly visible on outside	107		
Thread ends not trimmed to 1/4 inch or less on all seams with the exception of sleeve setting seams or loose threads not removed		207	
Visible basting thread not removed		208	
Series 500 stitch types trimmed to less than 1 inch or more than 1-1/4 inch on sleeve		209	
Component and assembly			
Any component part or required operation omitted or added (unless otherwise classified herein)	108		

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Component and assembly (cont'd)			
Any operation not performed as specified (unless otherwise classified herein)		210	
Cutting			
Any component part not cut in conformance with directional lines indicated on patterns or not cut in accordance with document requirements	109		
Seams and stitching			
Accuracy of seaming and stitching:			
Seam irregular or twisted, puckered 2/, pleated or wavy	110		
Part of material caught in unrelated operation or stitching:			
-caught in up to, and including 1/4 inch of stitching		211	
-caught in more than 1/4 inch of stitching	111		
End of stitching when not caught in other seams or stitching backtacked less than 1/4 inch			306
Stitching overlapping end of thread break less than 1/2 inch			307
Ends of a continuous line of stitching overlapped less than three stitches on labels or less than 1/2 inch on all other continuous stitching			308
Raw edge more than 1/4 inch long			309

2/ Puckering. In the course of the sampling examination, seams suspected of being puckered shall be examined at a distance of 3 feet in comparison with the AATCC photographic comparative rating (see 2.2) for seams. Puckering on a major portion of the suspected seam that equals or is worse than rating 3 for single needle seams shall be scored as a puckered seam.

Gage of stitching and seam allowance:			
Irregular, i.e., unevenly gaged or corresponding stitchings not uniformly gaged		212	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
Gage of stitching and seam allowance: (cont'd)			
Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			310
Open seam:			
On all seams except bottom hem and label stitching:			
Up to and including 1/2 inch			311
More than 1/2 inch up to and including 3/4 inch		213	
More than 3/4 inch	112		
Bottom hem more than 1/2 inch			312
Label stitching			313
Runoff:			
On joining seam, score as open seam			
Edge or raised stitching (when not resulting in an open seam):			
up to and including 1/2 inch			314
More than 1/2 inch		214	
Raw edge:			
On outside:			
Up to and including 3/8 inch			315
More than 3/8 inch		215	
On inside:			
More than 2 inches		216	
1/4 to 2 inches (inclusive)			316
Seam and stitch type:			
Not specified seam or stitch type		217	
Looper thread of 401 stitch type on outside	113		

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
Seam and stitch type: (cont'd)			
Any required line of stitching omitted	114		
Any line of stitching not beginning or ending where specified (unless otherwise classified herein)			317
Any seam not lapped or finished as specified		218	
Broken, missing, or skipped stitches on edge, top, or raised stitching (when seam is seamed, turned, and stitched) including edge, top or raised stitched darts			318
1/4 to 1/2 inch (inclusive)			319
More than 1/2 inch		219	
Stitch tension:			
Loose tension resulting in a loose seam	115		
Loose tension on raised or edge stitching resulting in exposed loose thread			320
Tight tension (stitches break when strain is applied in direction of seam or stitching)	116		
Stitches per inch (to be scored only when the condition exists on more than half the length of seam or stitching):			
Less than the minimum		220	
More than the maximum			321
Binding			
Raw edges not turned in			322
Raw edges not turned in and caught in stitching		221	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Buttonholes			
One or more omitted, added, misplaced, or not finished as specified	117		
Front end opening of horizontal top buttonhole not as specified		222	
End of buttonhole less than 3/8 or more than 7/16 inch from point of shoulder loop		223	
Shoulder loop buttonhole off-center by more than 1/8 inch		224	
Finished cut length not as specified		225	
One or more slanted		226	
Top buttonhole positioned vertically	118		
One or more of the lower five buttonholes positioned horizontally	119		
The lower five buttonholes unequally spaced by more than 1/4 inch			323
The lower five buttonholes less than 15/16 or more than 1-1/6 inches in from front edge		227	
Stitches not securely caught in fabric (stitches pull away from fabric)	120		
One or more broken, or more than two continuous skipped stitches in one or more buttonholes		228	
Loose stitch tension			324
Ragged edge in one or more			325
Not specified type or uncut			326
Tacking omitted on end		229	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Buttons			
One or more buttons missing, broken, or malformed	121		
Hand sewn button tacked off with less than two stitches		230	
Buttons not sewn through facing or insecurely sewn		231	
Stitching not locked at end of cycle (tug at loose end of thread to determine if it will ravel)		232	
Top shirt button (not collar button) not positioned as specified by pattern marks (see 6.8)	122		
Not positioned as specified		233	
Bartack or tack			
Missing	123		
Insecure, misplaced, not specified size or type, or not serving intended purpose		234	
Loose stitch tension			327
Labels			
Size label:			
Omitted, incorrect, illegible, or type IV label (when used) not completely on top collarstand	124		
Off center by more than 1/4 inch			328
Bottom of label less than 1/4 or more than 1/2 inch from neck/collarstand joining seam			329
Combination identification and instruction label:			
Omitted, incorrect, or illegible		235	
Not positioned on lower end of left facing		236	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Labels (cont'd)			
Combination identification and instruction label: (cont'd)			
Bottom of label positioned less than 3 or more than 3-1/4 inches from finished bottom			330
Edge of label less than 1 or more than 1-1/8 inch from finished shirt front			331
Darts			
Front or back darts not uniform in length by more than 1/4 inch	237		
Any dart tapered less than 1/2 or more than 5/8 inch beyond pattern mark			332
Not properly positioned		238	
Has pleats, puckers, or gathers		239	
Point(s) not tacked, backtacked, or backstitched		240	
Collar			
Not uniform in size or shape - any point varying:			
More than 3/8 inch with corresponding point on matching part	125		
1/4 to 3/8 (inclusive) inch with corresponding point on matching part		241	
Collar point uneven in length by:			
1/8 to 1/4 (inclusive) inch			333
More than 1/4 and up to 3/8 (inclusive) inch		242	
Over 3/8 inch	126		

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Collar (cont'd)			
Twisted, not smooth, too full, short, or tight causing collar to turn out	127		
Collar point not properly forced out	128		
Collar off center by 3/:			
More than 3/16 and up to 1/4 (inclusive) inch			334
Over 1/4 inch		243	
Pucker or pleat at neckline		244	
Edge not worked out (having folds 1/16 inch or more deep)			335
Edge of undercollar exposed beyond edge of top collar:			
1/16 inch or more for a distance of 2 inches or more	129		
Less than 1/16 inch on major portion of collar		245	
One or both collar stays missing	130		
One or both stays not attached or positioned as specified	131		
End of collarstand finished more than 1/8 inch beyond front edge		246	
Collar ends open less than 1/4 or more than 1/2 inch when buttoned		247	
Width of collarstand varying more than 3/16 inch when measured at the two shoulder seams at junction of collar			336
Tunnel collar stay stitching missing		248	
Top and sides of collar not edge stitched as specified		249	
Not placed between undercollar parts as specified		250	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Collar (cont'd)			
Edge of collarstand not turned under and caught in stitching			337
Tunnel collar stay stitching missing			338
Top and sides of collar not edge stitched as specified			339
3/ Center of collar is determined by aligning the two shoulder seams at junction of collar			
Shoulder loop			
Not uniform in size or shape - any point varying:			
More than 3/8 inch with corresponding point on matching part	132		
3/8 inch or less with corresponding point on matching part		251	
Not positioned as specified or attached crookedly:			
Point of loop more than 1/8 inch from collar setting seam		252	
Front edge of loop at armhole opening less than 15/16 or more than 1-1/16 inch forward of shoulder seam		253	
Not of the specified dimension:			
Less or more than specified dimension by 1/8 inch		254	
Less or more than specified dimension by 1/4 inch	133		
Loop tight, loose, or badly distorted when buttoned		255	
Points and edges not worked out			340
Shoulder area distorted when shoulder loop is buttoned causing fabric pleats or tucks in shirt		256	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Front			
Front tight or short, or front facing tight or twisted causing noticeable bulge or twist when shirt is buttoned	134		
Uneven in length at bottom when buttoned, by more than 1/2 inch			341
Front facing			
Facing not caught as specified			342
Back side of facing not overedged stitched		257	
Width of facing not as specified			343
Shoulder end of facing not attached to shoulder seam edge as specified			344
Armhole and sleeve			
Sleeve tight causing pleat or excessive puckers on front, back, or sleeve		258	
Sleeve pleats not as specified		259	
Sleeves reversed	135		
Construction of sleeve opening:			
End of binding not securely caught in sleeve opening i.e., can be easily pulled away	136		
Bound edge not securely stitched in cuff		260	
Binding twisted or puckered		261	
Binding not finishing 3/8 ± 1/16 inch wide		262	
Sleeve opening less than 2-3/4 or more than 3-1/4 inches long along outside fold exclusive of cuff		263	

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Armhole and sleeve (cont'd)			
Construction of sleeve opening: (cont'd)			
Sides of sleeve opening uneven in length by 1/4 inch or more (measured to top edge of cuff)		264	
Raw edges of binding not turned in			345
Binding not stitched or bartacked 1/8 to 1/4 inch below top of sleeve opening			346
Cuffs			
Construction and setting:			
Reversed, i.e., buttonhole on underside and button on top side	137		
End of cuff finished beyond outer edge of sleeve opening by more than 1/8 inch		265	
Width not within specified tolerance:			
Up to 1/8 inch			347
More than 1/8 inch		266	
Irregular in width by 1/4 inch or more		267	
Edge stitching on cuff not as specified			348
Buttonhole misplaced or not positioned as specified	138		
Looper side of 401 stitch when used not finishing on underside of cuff		268	
Joining of front to side			
Not as specified			349
Joining of back to fronts			
Not as specified			350

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TABLE V. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Bottom hem			
Not as specified			351
Width not as specified			352
Hem twisted, pleated, or puckered		269	
Neck fastener loop tapes			
Less than 9/16 or more than 11/16 inches long			353
Not color specified			354
Positioned less than 1-5/16 or more than 1-7/16 inches from collar edge		270	
Fusible material			
Omitted in top collar, front facing, cuffs, shoulder loops and collarstand	139		
Delamination: Partial or complete delamination on any fused component	140		
Bubbling on any fused component	141		
Any strike through or bleed through	142		
Any resin transfer to shirt	143		

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table III. Any dimension not within the specified tolerance or sleeves uneven in length by more than 1/2 inch shall be classified as a defect. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table V. The lot size shall be expressed in units of shirts. The sample unit shall be two shirts. The sample size shall be as specified in 4.4.1.1. One shirt shall be used to evaluate conformance to the appearance requirements and the other shirt shall be used to evaluate for bubbling and

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delamination of the collar and delamination of the collar stays. The lot shall be acceptable if one or more sample units fail to meet the requirements for fused collars. Additionally, the lot shall be unacceptable if the lot average (average of all the determinations on the sample units for each characteristic) fails to meet any specified requirement.

TABLE VII. End item tests

Characteristic	Requirement paragraph	Test method
Seam appearance rating	3.7.1	4.5.1.1
Fusibles resistance to laundering	3.7.2	4.5.1.1 1/
Ultrasonically fused collar stay resistance to laundering	3.7.3	4.5.1.1 1/

1/ After laundering and conditioning, the fusibles shall be visually examined for conformance to 3.7.2 and then, when applicable, the collars shall be opened and the stays shall be examined for conformance to 3.7.3.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: Incomplete sealing or closure of flaps, loose strapping, improper taping, or inadequate stapling Bulged or distorted container
Content	Number of shirts per container is more or less than specified 1/

1/ For this defect, one shipping container in the sample shall be examined.

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4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Seam appearance rating and fusible resistance to laundering test.

4.5.1.1 Laundering procedure.

4.5.1.1.1 Apparatus and material used for laundering. The following apparatus and material is required for laundering:

a. Automatic washing machine (see 6.6).
b. Automatic dryer (see 6.6).
c. AATCC Standard Detergent 124 or equivalent.

4.5.1.1.2 Laundering and conditioning. If the shirts or materials selected as test samples weigh less than 4 pounds, ballast to make up a standard 4-pound load shall be added to the test samples. The ballast shall consist of durable press shirts that are approximately the same weight as the test shirts or pieces of durable press fabric that are similar to the fabric of the test samples. The pieces shall be hemmed and measure 3 feet by 3 feet. The front of all shirts in the 4-pound load shall be unbuttoned. The load shall be placed in the washing machine and 90 grams of detergent shall be added to the washing machine. The washing machine shall be set for 10 minutes on the wash and wear (permanent press) phase, high water level. The washing machine shall be filled with water with a hardness not greater than 50 parts per million (PPM), and the wash water shall be at a temperature of 115° ± 5°F with a cold rinse. Immediately after the washing machine has completed the final spin phase, the entire load shall be removed, untangled, and placed in the dryer. The dryer shall be operated at a temperature of 140° to 160°F until the entire load is dry and then tumbled for an additional 10 minutes with the heat turned off (cool-down phase). The load shall be removed immediately after the machine stops. The above washing and

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drying procedure constitutes one complete cycle. Test sample material for adhesion testing shall be subjected to 5 complete cycles. Test sample shirts for collar fusibles and collar stay evaluation shall be subjected to 10 complete cycles. After completion of the last cycle and prior to evaluation, the test sample shall be conditioned for 2 hours at 70° ± 10°F and at a relative humidity of 50 to 65 percent. (NOTE: During the interruptions between the cycles and during the conditioning and the evaluations, test shirts shall be placed on wooden or plastic hangers and shall be buttoned at the front.)

4.5.1.2 Seam appearance evaluation.

4.5.1.2.1 Apparatus and material used for evaluation. The following apparatus and material is required for evaluation of samples for seam appearance:

a. Overhead lighting procedure apparatus as specified in AATCC Test Method 124-78.

b. AATCC photographic standards for evaluating single needle stitching.

c. Evaluation room where the only source of light will be that used for observation. To prevent unwanted reflection of light that will interfere with the observation, the walls of the room may be painted a flat non-reflecting black, or blackout curtains may be hung on each side of the viewing area.

4.5.1.2.2 Evaluation of the appearance of seams. After the test shirts have been laundered and conditioned as specified in 4.5.1.2, the shirts shall be evaluated by three trained observers. The trained observers, independent of the other observers, shall evaluate each of the shirts as follows: The shirt shall be placed on a clothes hanger with the buttons fastened and the hanger suspended on the viewing board in such a manner that the shirt lies flat on the surface of the board. The Method 88B photographic standards for single needle stitching shall be placed on the viewing board. The observer shall stand directly in front of but 4 feet from the test shirt. The observer shall then compare the test shirt, in each of the areas specified in 3.7, to the photographic standards and shall assign a rating value to each of the specified areas. The rating value assigned to an area by the observer shall be the numerical value of the photographic standard that most nearly matches the area. Seams appearing to fall between the numerical values of the photographic standards may be rated 1.5, 2.5, 3.5 or 4.5. Each rating value assigned by each of the observers shall be recorded. The average rating values assigned by each observer to the same area shall be the rating value for that area. If the rating of the same area made by two observers differs by more than 1.0, the rating of all three observers made for that area shall be discarded and the area shall be re-evaluated by three different trained observers.

4.5.1.2.3 Bubbling and delamination of collars. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2, the collars shall be examined visually for bubbling and delamination.

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4.5.1.2.4 Adhesion of ultrasonically fused collar stays. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2 and collars examined for bubbling and delamination as specified in 4.5.1.2.3. The collar shall be opened and the stays examined visually for delamination and curling.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each fully buttoned shirt shall be neatly folded to measure approximately 13 inches in length by 10 inches in width. The folded shirt shall be secured with stainless steel (AISI type 304 or 316, see 2.2) pins, aluminum clips or plastic fasteners. Each folded shirt shall be unit packed in a close fitting, clear polyethylene film bag conforming to A-A-50083.

5.1.2 Commercial preservation. The shirts shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Forty-six shirts of one type and size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width and twenty-three in depth. Inside dimensions of each shipping container shall approximate 20-1/2 inches in length, 13-1/2 inches in width, and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II, water-proofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Forty-six shirts of one type and size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and twenty-three in depth. Inside dimensions of each container shall approximate 20-1/2 inches in length, 13-1/2 inches in width, and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II as specified in PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

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5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial. Shirts, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951, as applicable.

5.3 Palletization. When specified (see 6.2), shirts, packed as specified in 5.2.2 or 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups, I, II, III, or IV, of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 7 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

6.1 Intended use. The long sleeve shirts are intended for wear by female military personnel of the Department of the Army.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Type and size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 Samples. For access to samples address the contracting activity issuing the invitation for bids or request for proposal.

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6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.5 Seam appearance. In order to avoid puckering and to produce the flattest seams possible, consideration must be given to the following:

a. The needles used for all seams should be as fine as possible consistent with the size or diameter of the thread used.

b. The throat plate hole should be no larger than necessary to permit free entry of the needle and to prevent flagging.

c. The finest feed (approximately 18 teeth) possible should be used with a pressure foot to exactly match the sample and size of the feed. The face of the pressure foot (underside next to the feed on the machine) should be highly polished to eliminate roughness and reduce drag. Diamond shaped feed teeth have been used quite successfully.

d. The lightest pressure foot pressure should be used to prevent stretching of the top ply of material.

e. In the safety stitch machine (515, 516 and 519), the upper and lower trimming knives should be kept at peak sharpness to prevent "dragging" of the end of the fabric.

f. In stitch type 301 sewing, the bobbin tension should be adjusted so that when a full bobbin is placed in the bobbin case and threaded through the tension spring, the bobbin and case shall ride slowly down the thread when the free end of the thread is held in the vertical position. Needle tension should be adjusted after the bobbin tension to provide slightly more needle thread than bobbin thread in the stitch.

g. When sewing with stitch type 401 (including the seaming row of the safety stitches), the thread ratio (needle and looper) should be adjusted to produce a stitch with 60 to 70 percent needle thread, and both needle and looper thread tension should be held to the minimum.

h. Thread takeup mechanism should be set on all machines to relieve as much tension as possible when forming the stitch. All thread contacts should be well polished and free of burrs, cuts and rough spots.

6.6 Apparatus. The recommended apparatus for 4.5.1.1.a. and b., and for pressing in operation 26 of table II are:

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a. Kenmore Automatic Washing Machine Model 600, or similar washing machine. In case of dispute, the Kenmore Model in use at the procuring agency laboratory.

b. Kenmore Automatic Dryer Model 600, or equivalent.

c. "Formatic" hot head pressing machine or equal. "Formatic" manufactured by the American Laundry Machine Co., Cincinnati, OH.

6.7 Defect scale. Sears Fabric Defect Scales are available from:

Sears Roebuck Company
Department 817 (Attn: BSC 23-29)
"Sears Defect Replica Kit"
Sears Tower
Chicago, IL 60684

6.8 Placement of top shirt button. Distance from collarstand setting seam to center of top shirt button should measure 5-3/4 ± 1/16 inch for size 12. (Measurement will change respective to size.) Any less results in top shirt button being exposed when worn with the coat, classic.

6.9 Use of automated apparel equipment. The automatic creaseband construction is the crease and tack process from Facemate Corporation.

6.10 Subject term (keyword) listing.

Clothing, women's
Warm weather

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8410-A759)

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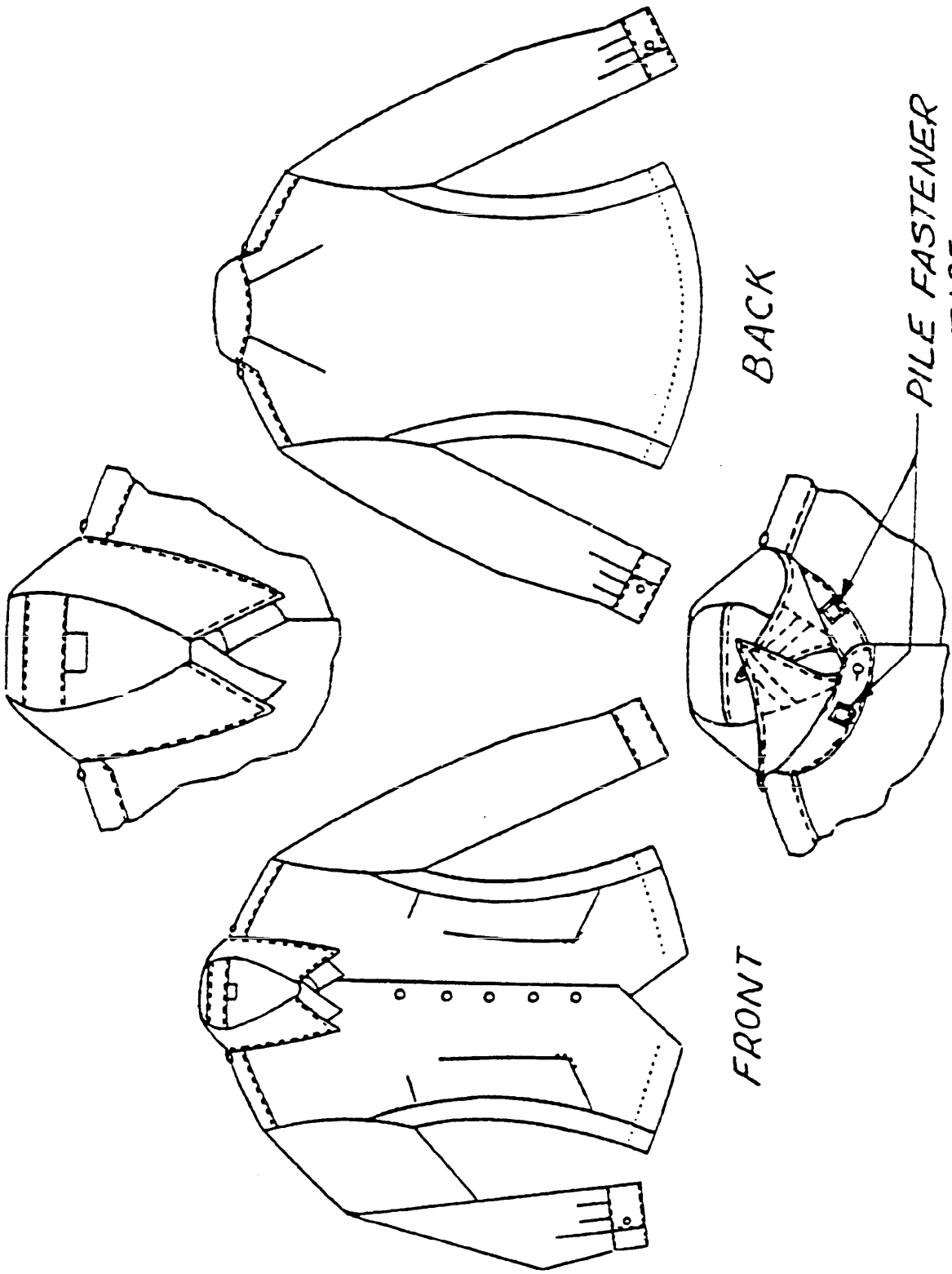


FIG. 1-SHIRT, WOMAN'S, LONG SLEEVE, POLYESTER/COTTON, ARMY GREEN, 415, DURABLE PRESS 2-1-1532

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-44093B(GL)	1990 November 8

3. DOCUMENT TITLE
SHIRT, WOMAN'S, LONG SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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